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Paleomagnetism of the Carboniferous Tepuel Group, central Patagonia, Argentina

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Abstract. Samples of the Carboniferous glaciogenic sequences of the Tepuel Group, western central Patagonia, Argentina, were collected from 23 sites in two localities, Sierra de Tepuel (43.7°S, 70.5°W) and Las Salinas (43.2°S, 70.3°W). Most were fine sandstones to siltstones of late Visean to Westphalian age. A characteristic remanent magnetization was determined in most sites by alternating field and/or thermal cleaning. At Sierra de Tepuel all samples were partially overprinted by a secondary magnetization apparently related to Mesozoic magmatic activity in this area. However, site mean directions of the characteristic remanence from both localities passed a tilt test, indicating a pretectonic (pre-Early Permian?) magnetization. Isothermal remanent magnetization experiments showed magnetite as the main and often the only magnetic carrier, while thin section analyses indicated a detrital origin for this mineral, suggesting a depositional or postdepositional magnetization. The paleomagnetic pole for the Tepuel Group (Namurian-Westphalian) at 316.1°E, 31.7°S (number of sites N=16, minor and major semi-axes of confidence $\delta p = 15.0°$, $\delta m = 16.0°$), is considered a key pole for the Middle Carboniferous of South America and Gondwana, confirming the apparent polar wander path for the supercontinent for which the pole moved from central to southern Africa in the Early Carboniferous, followed by a rapid motion from southern Africa to Antarctica during the late Westphalian. The agreement of the Tepuel pole with this Gondwanan path confirms that Patagonia was already part of South America by the Middle Carboniferous.

Introduction

A well-defined apparent polar wander path (APWP) is of fundamental importance when assessing the paleogeographic and geodynamic evolution of any continent. Unfortunately, there are few reliable Paleozoic paleomagnetic poles (PP) for South America, and most of these are of latest Carboniferous or Permian age [e.g., *Vilas*, 1981]. Consequently, the South American Paleozoic APWP has major gaps and is largely dependent on paleomagnetic data from other Gondwanan continents for assessing its paleogeographic and geodynamic evolution throughout the Paleozoic. However, even the Paleozoic APWP for the whole of Gondwana is a matter of controversy, with very different paths recently proposed [e.g., *Bachtadse and Briden*, 1990, 1991; *Li et al.*, 1990]. These are also dependent on the validity of the Gondwanan reconstruction adopted [e.g., *Smith and Hallam*, 1970; *Lawver and Scotese*, 1987; *Tarling*, 1988]. Thus high-quality Paleozoic PPs are required for South America, as well as other Gondwanan continents, in order to define their paleogeographic and tectonic evolution.

Some authors [*Dalmayrac et al.*, 1980; *Ramos*, 1984; 1988; *Kay et al*, 1989] have proposed that large parts of southwestern South America are allochtonous and sometime during the Paleozoic were accreted, either as a single continental mass or as separate terranes, onto the Precambrian South American Shields [*Ramos*, 1988]. One of the largest of these blocks is that of Patagonia, which, according to *Ramos* [1984; 1988], collided with the Rio de La Plata Craton in the late Paleozoic. According to this model, the late Paleozoic Ventana and Cape Fold Belts of Argentina and South Africa, respectively, would be the result of the collision of the Patagonian land mass against Gondwana (see also *Daly et al.* [1991]). However, there is no undisputed evidence of the allochthonous nature of Patagonia as a whole [see *Dalziel and Grunow*, 1992]. Few paleomagnetic data have yet been produced to evaluate these hypotheses. Recent results from Silurian-Devonian sediments in northeast Patagonia [*Rapalini and Vilas*, 1991] suggest no differential movements between Patagonia and Gondwana since the Early Devonian, but no definite conclusion could yet be drawn owing to the low number of samples analyzed and the largely undefined nature of the mid-Paleozoic section of the Gondwanan APWP.

In order to improve the South American (and therefore Gondwanan) APWP and to constrain the tectonic evolution of Patagonia during the late Paleozoic a paleomagnetic study was carried out on the thick Carboniferous glaciogenic sequences of the Tepuel Group in western central Patagonia, Argentina. Such a study was also considered important to further assess the paleolatitude of such glacial deposits.

Geologic Setting and Sampling

Late Paleozoic deposits (Figure 1) make up the fill of the Central Patagonian Basin (also called Tepuel-Genoa) in western central Patagonia [*Andreis et al.*, 1987]. These deposits were first dated by *Suero* [1948]. Since then numerous studies followed, mainly by Argentine geologists

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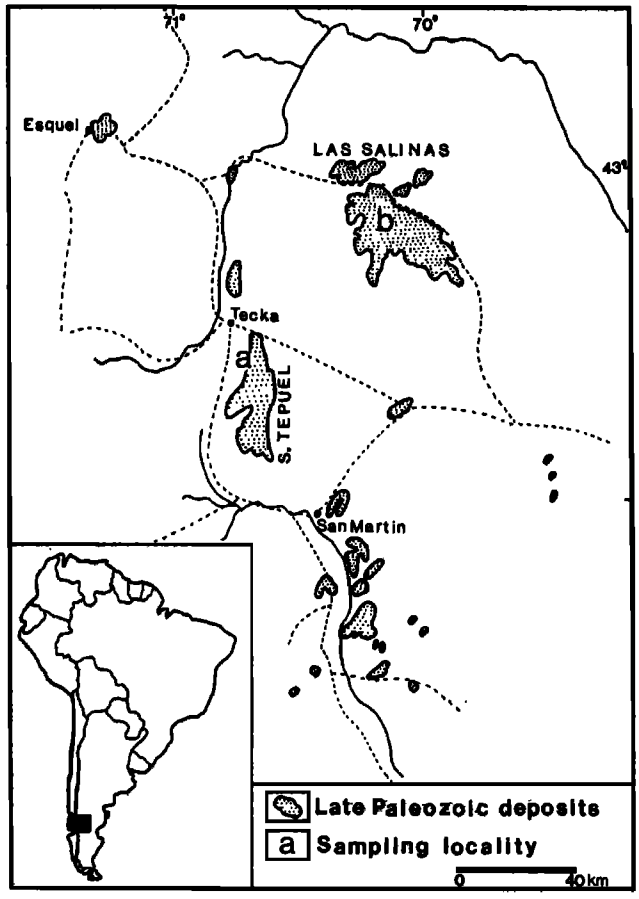


Figure 1. Sketch of exposures of late Paleozoic deposits in western Central Patagonia and sampling localities a and b.

(see review by *Andreis et al.* [1987]). The stratigraphic record of the basin spans Early Carboniferous to Late Permian and has been divided into two main units, the Carboniferous Tepuel Group [*Suero*, 1948] and the Permian Rio Genoa Group [*Lesta and Ferello*, 1972]. While the former, mainly exposed in the north of the region, is basically composed of marine and in parts, glaciogenic deposits, the Rio Genoa Group largely comprises continental or littoral sediments and is exposed in the southern parts of the basin. The glaciogenic character of certain levels of the Carboniferous deposits in the Central Patagonian Basin was determined long ago [*Suero*, 1948] and has since attracted many detailed sedimentologic, paleogeographic, and paleoclimatologic studies [e.g., *Frakes et al.*, 1969; *Frakes and Crowell*, 1969; *Crowell and Frakes*, 1975; *Gonzalez*, 1984; *Lopez Gamundi and Limarino*, 1984; *Lopez Gamundi*, 1989; *Gonzalez Bonorino*, 1992]. The age of these sequences has been accurately determined by their fossil content [see *Andreis et al.*, 1987].

The most complete sequence of the Carboniferous rocks (the Tepuel Group) is exposed at Sierra de Tepuel (Figure 1, locality a), with about 4000 m of sandstone, siltstone, claystone, and diamictites, and at Sierra de Languiñeo, near Las Salinas (Figure 1, locality b), where its thickness is estimated to be 2500 m. Both sequences are roughly equivalent in age, although the oldest levels at Sierra de Tepuel seem to be absent at Las Salinas [*Gonzalez*, 1972; *Andreis et al.*, 1987]. Further details on the geology of both areas are given by *Suero* [1948], *Gonzalez* [1972], and *Page et al.* [1984], among others.

On the western slope of Sierra de Tepuel the sequence has been tilted to the east and is virtually a homoclinal structure which, according to *Suero* [1948], is the eastern limb of a large N-S anticline, the axis and western limb of which are not exposed. The age of the tectonic movements is not well constrained but is definetly post Early Jurassic, as Liassic deposits conformably overlie the Carboniferous sequence [*Gonzalez*, 1984]. The sequence is an exhumed marine basin to shelf system which has been divided into three formations [*Page et al.*, 1984], the Jaramillo, the Pampa de Tepuel and the Mojón de Hierro Formations. Jurassic gabbroic dykes of the Cresta de los Bosques Complex intrude the Pampa de Tepuel Formation in this locality [*Page*, 1984].

1. The Jaramillo Formation is composed of nearly 1000 m of fine sandstone and claystone assigned to coastal and littoral environments. Its age, on the basis of fossil plants [e.g., *Petriella and Arrondo*, 1978], is Early Carboniferous (Visean). Block samples were collected at eight sites in this formation (GT1-GT8).

2. The Pampa de Tepuel Formation consists of 2900 m of marine claystone, siltstone, sandstone, and some conglomerates and limestone and has at least three main glacial diamictite levels [*Gonzalez Bonorino*, 1992]. Numerous marine invertebrates belonging to the *Levipustula levis* zone indicate a Namurian-early Westphalian age [*Riccardi and Sabattini*, 1975; *Lopez Gamundi*, 1989]. Six sites (GT8-GT13) were drilled in the lower levels of this formation, immediately above Glacial Horizon 1 [*Gonzalez Bonorino*, 1992]. A detailed sedimentological log (Figure 2a) was made of this section. It consists of a lower, basinal marine shale unit (unit 1), characterized by the presence of dropstone beds and diamictites, which have been interpreted by previous workers as turbidite debris flows related to glaciation activities [*Lopez Gamundi and Limarino*, 1984; *Lopez Gamundi*, 1989] or as rain-out till [*Gonzalez Bonorino*, 1992]. Overlying this is a 60-m thick sequence (unit 2) of channel fill and normally graded sheet sandstones beds [Figure 2a]. The sandstones contain shale rip-up clasts and are characterized by erosional bases and a massive lower half to each bed. The upper parts of the beds exhibit parallel lamination and have current rippled tops. These vertical trends in sedimentary structures probably represent incompletely developed Bouma sequences, related to deposition from unidirectional, turbulent flow. The channel/sheet sandstone system fines upward into several hundred meters of silty shales (unit 3). Within this undeformed unit is a series of spectacularly synsedimentary deformed sandstone units up to 30 m thick. A series of these units, at a common stratigraphic horizon, can be mapped out along a 10-km transect. From our observations, unit 3 is interpreted as an unstable slope with large slump units, probably triggered by oversupply of sediment during deglaciation.

3. The Mojón de Hierro Formation is composed of 500 m of littoral to fluvial quartzites and massive claystones and has been assigned to the Late Carboniferous on the basis of

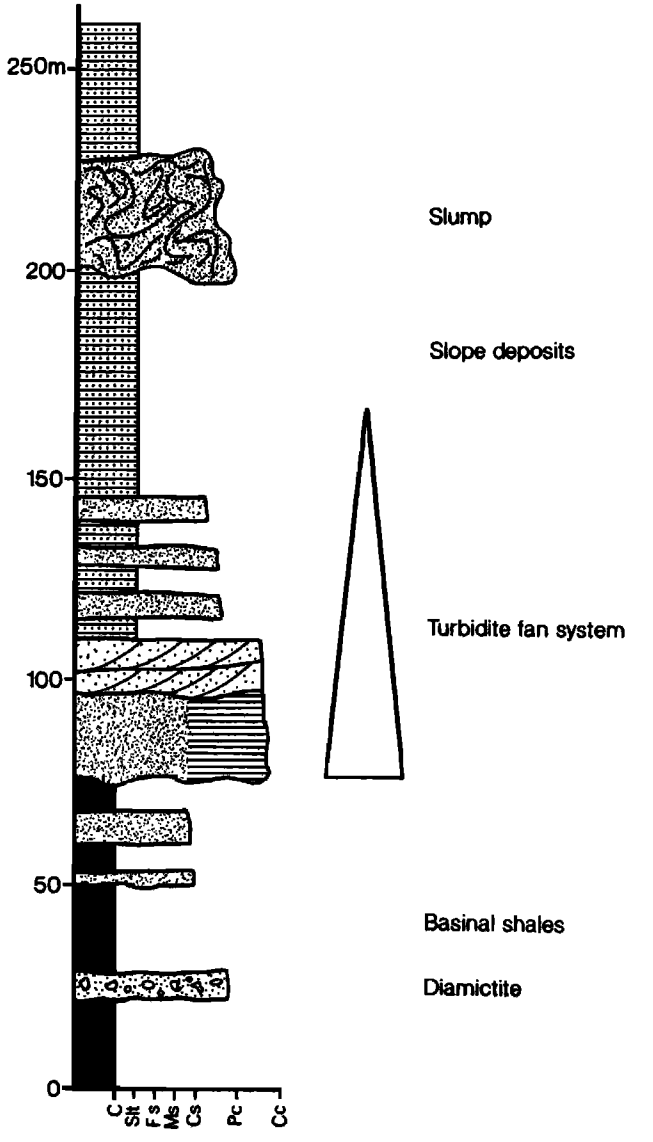


Figure 2a. Summary sedimentological log of units 1-3 in the Pampa de Tepuel Formation, Sierra de Tepuel. Note the abrupt, erosive base to the turbidite channel system and the fining upward into the slope unit. At least 10 deformed, slumped sandstone units occur along strike at the same depositional level.

its marine invertebrates and fossil plants [e.g., *Andreis and Cúneo*, 1985; *Gonzalez*, 1985]. This formation was not sampled at this locality.

In Las Salinas the Carboniferous sequence has been folded in an open anticline plunging 30°NNE and intruded by a posttectonic granodiorite that is exposed in the southern tip of the outcrops [*Gonzalez*, 1972]. This author and *Stipanicic and Linares* [1969] suggested an Early to Late Permian age for the granodiorite, by correlation with similar intrusives to the north of the Las Salinas area which have given those radiometric ages. In such a case the age of the deformation in locality B would be constrained to the latest Carboniferous-Early Permian. However, the single radiometric dating obtained on these rocks up to now yielded 33 ± 2 Ma [*Spikermann*, 1978]. Despite not being considered as a reliable indication of the true age of the intrusive by the proper author, it rises doubts about the validity of the proposed Permian age. The sedimentary sequence corresponding to the Tepuel Group at Las Salinas is divided into nine members [*Gonzalez*, 1972], LS-1 to LS-9, the first five (or six) apparently corresponding to the Pampa de Tepuel Formation and the last four (or three) to the Mojón de Hierro Formation. The stratigraphy of the succession at this locality is probably complicated by a series of SSW directed thrusts which possibly repeat some beds. Six sites (GT14-GT19) were block sampled in members LS5 and LS6, which are a mixed continental, shallow marine, and glaciogenic package. These were logged sedimentologically (Figure 2b) at Cerro Antinao where grey siltstones coarsen upward into sandy heterolithic material and

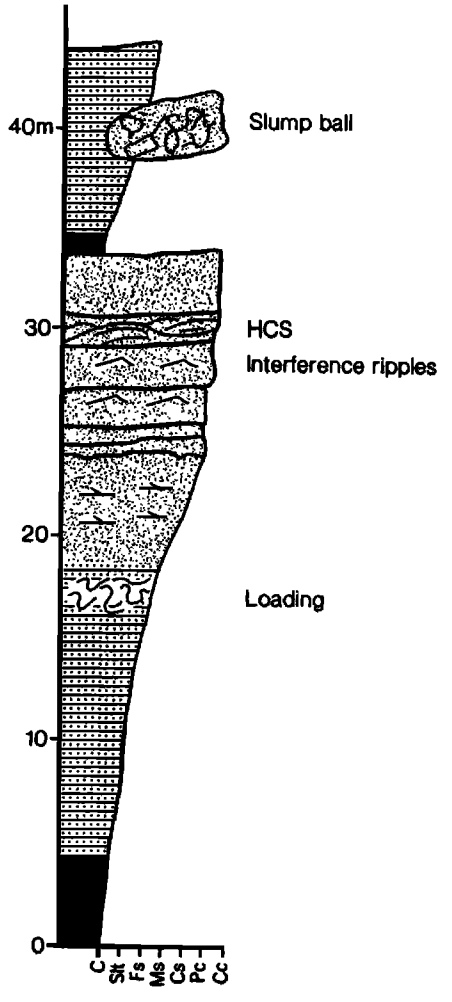


Figure 2b. Detail of two progradational, coarsening upward units from the Tepuel Group at Las Salinas. These units are interpreted as mixed current/storm-dominated, wave-influenced deltaic mouth bars. The flooding surface at the top of the lower unit represents an abrupt rise in relative sea level. The slumped sandstone bed represents oversupply of sediment and subsequent failure of the delta front, probably related to exceptional floods. HCS is hummocky cross stratification.

eventually into amalgamated sheet sandstone beds characterized by hummocky cross stratification. The heteroliths show loading and interference ripples, indicative of high rates of deposition and the interaction of current and wave processes. This association of facies suggests that this unit is a prograding storm/wave-influenced mouth bar. Several of these 30 to 40-m thick coarsening upward units are vertically stacked, separated by abrupt flooding surfaces.

The younger sequence block sampled at Las Salinas was member LS-9 (four sites, GT-20 to GT-23) which included coarse-grained sandstones with channel geometries, associated with thin sheet sandstones and grey/red siltstones, presumably representing a semiarid alluvial channel/floodplain environment.

Four to seven drilled cores or three to four block samples (from which at least two cores were subsequently drilled in the laboratory) were collected at each site.

Paleomagnetic results

From each 2.5-cm-diameter drill core, generally two specimens, each 2.2 cm high, were cut. Two to three pilot specimens were selected from each site and submitted to stepwise magnetic cleaning, both alternating field (AF) and thermal. Most measurements were made with a 2G cryogenic magnetometer at the University of Southampton. All samples showed low values of initial remanence intensity, generally less than 0.5 mA/m. AF demagnetization was performed on at least one pilot specimen per site in 10 to 12 stages up to magnetic fields of 60 mT. Pilot thermal demagnetization was also performed on at least one pilot specimen per site, generally in nine steps (150°, 250°, 300°, 350°, 400°, 450°, 500°, 530°, 560°, and 580°C). During thermal cleaning of the pilot samples the bulk susceptibility after each demagnetization step was checked, in order to monitor the occurrence of chemical changes associated with experimental heating. All the remaining specimens were submitted to five or six demagnetization steps using the most effective demagnetization technique and ranges for each site, as assessed from the pilot study. This meant that the remaining samples from Sierra de Tepuel were thermally demagnetized, while those from Las Salinas were submitted to AF demagnetization, as the thermal method had not proved more effective and AF methods are quicker and do not involve thermally induced chemical changes.

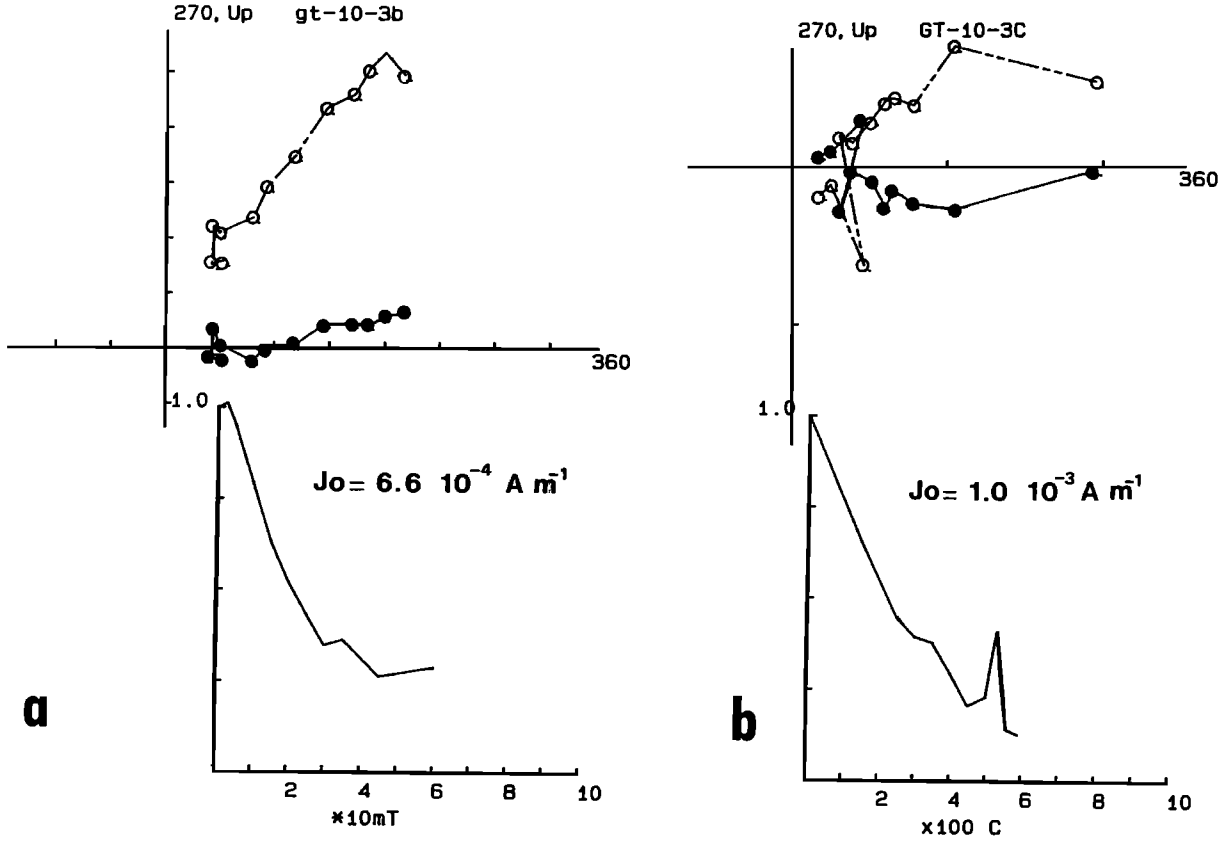


Figure 3. (a) Typical magnetic behavior of a sample from the Tepuel Group at Sierra de Tepuel submitted to alternating field demagnetization. Open circles in the Zijderveld diagram (top) indicate projections on the vertical planes; solid circles, horizontal planes. Numbers on each axis indicate geographic azimuth. Normalized intensity decay after each demagnetization step is shown in the Cartesian plot (bottom). Jo is natural remanent magnetization (NRM) intensity in amperes per meter. (b-d) Typical magnetic behaviors of samples from the Tepuel Group submitted to thermal demagnetization. Symbols are same as in Figure 3a.

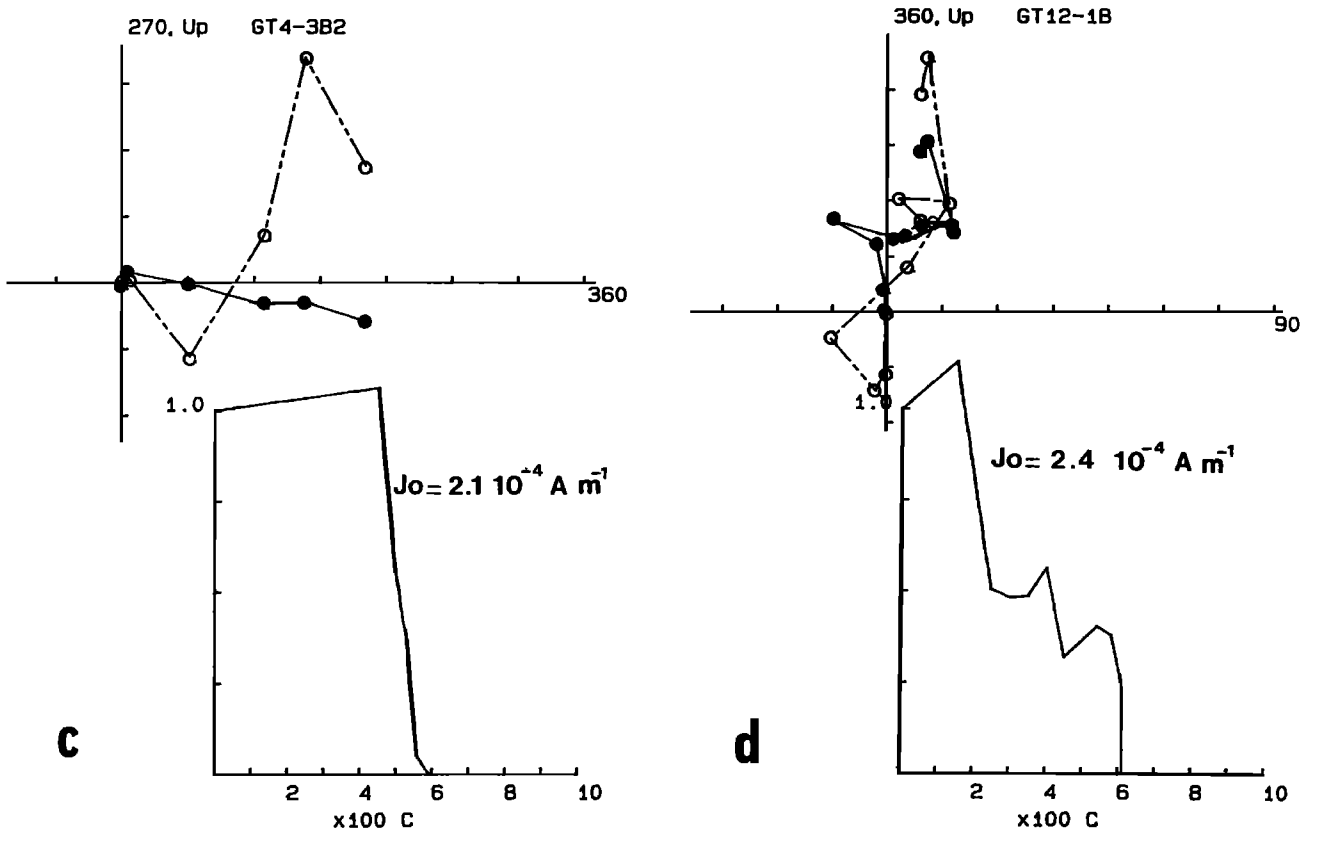


Figure 3. (continued)

Samples from Sierra de Tepuel generally carried two magnetic components. A dominant north and upward directed component (in situ coordinates), with moderate to steep inclination (component A), could be isolated in most samples by incremental AF demagnetization (Figure 3a) and also by thermal demagnetization up to 500°C (Figure 3b). The second component, directed north with a moderate downward inclination (component B), was observed in most sites by thermal demagnetization over 500°C (Figures 3b-3d) but was not discernible with the AF method. This suggests that the coercivity spectrum of component B is completely overlapped by that of component A, while both components blocking temperature spectra are only partially overlapped. Component B was detectable in most samples, but it could only be defined in less than half of them, as it became obscured by magnetic changes associated with thermochemical reactions in many samples at about 550°C. Both components seem to be carried by magnetite, as neither magnetization survived heating above 580°C.

Samples from Las Salinas showed only one characteristic magnetization, defined by AF cleaning (Figures 4a-4c), which was directed south and downward (in situ coordinates). Only one sample from site 18 presented an antipodal direction (north, upward). This was commonly overprinted by a comparatively high-intensity soft (viscous?) magnetization with no consistent orientation (Figures 4a and 4b). This and the presence of some magnetically unstable samples in this locality suggest that multidomain (MD) magnetic grains are common in Las Salinas, while the broad spectra of blocking temperatures probably reflect variable sizes of magnetic domains (Figure 5). Thermal demagnetization shows complete deletion of magnetization around 580°C, which is thought to reflect the Curie temperature of magnetite, probably the magnetic carrier in these samples.

Mean directions of magnetization for each site were computed for component A at Sierra de Tepuel (Table 1). Sample directions and site mean directions for component B from Sierra de Tepuel and the characteristic remanence of Las Salinas are presented in Tables 2 and 3. The low consistency of these latter directions within a given site contrasts with a good consistency between sites. This may be due to random components associated with MD (and possibly superparamagnetic (SP), see Figure 9a) grains that tend to obscure the characteristic remanence; note that very large but soft random viscous components were frequently observed in these rocks (Figures 4a and 4b). The difficulties in accurately defining the very weak, high-temperature component B, often affected by thermochemical changes during heating, also contribute to the low consistency of directions within a site. However, such good consistency between sites supports the overall validity of the site mean values.

Rock Magnetic Properties

In order to identify the minerals carrying the remanence, which is vital in determining its origin, isothermal

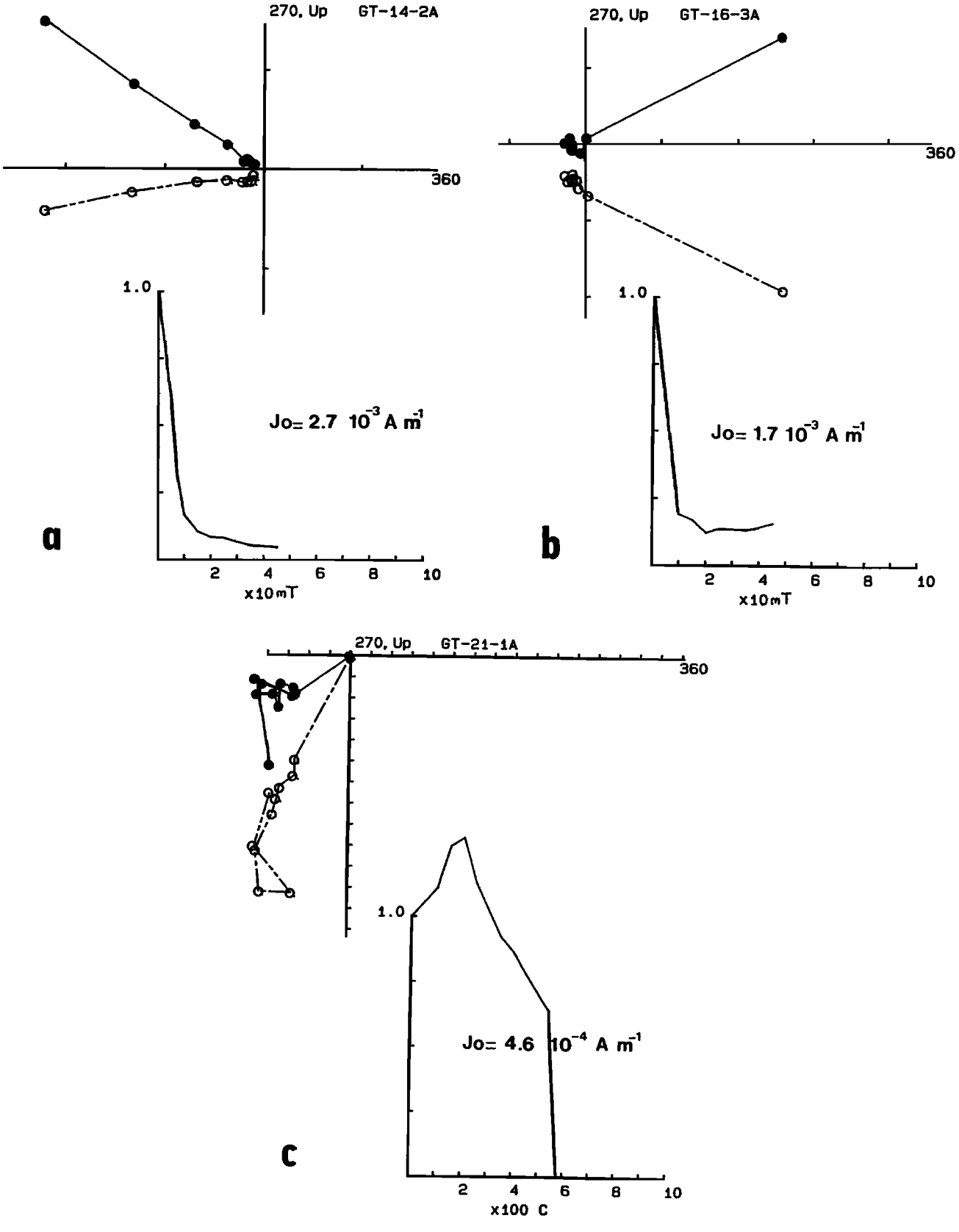


Figure 4. Typical magnetic behavior of (a) sample GT14-2A and (b) sample GT16-3A from the Tepuel Group at Las Salinas submitted to AF demagnetization. (c) Magnetic behavior of sample GT21-1A from the Tepuel Group at Las Salinas submitted to thermal demagnetization. Symbols are same as in Figure 3a.

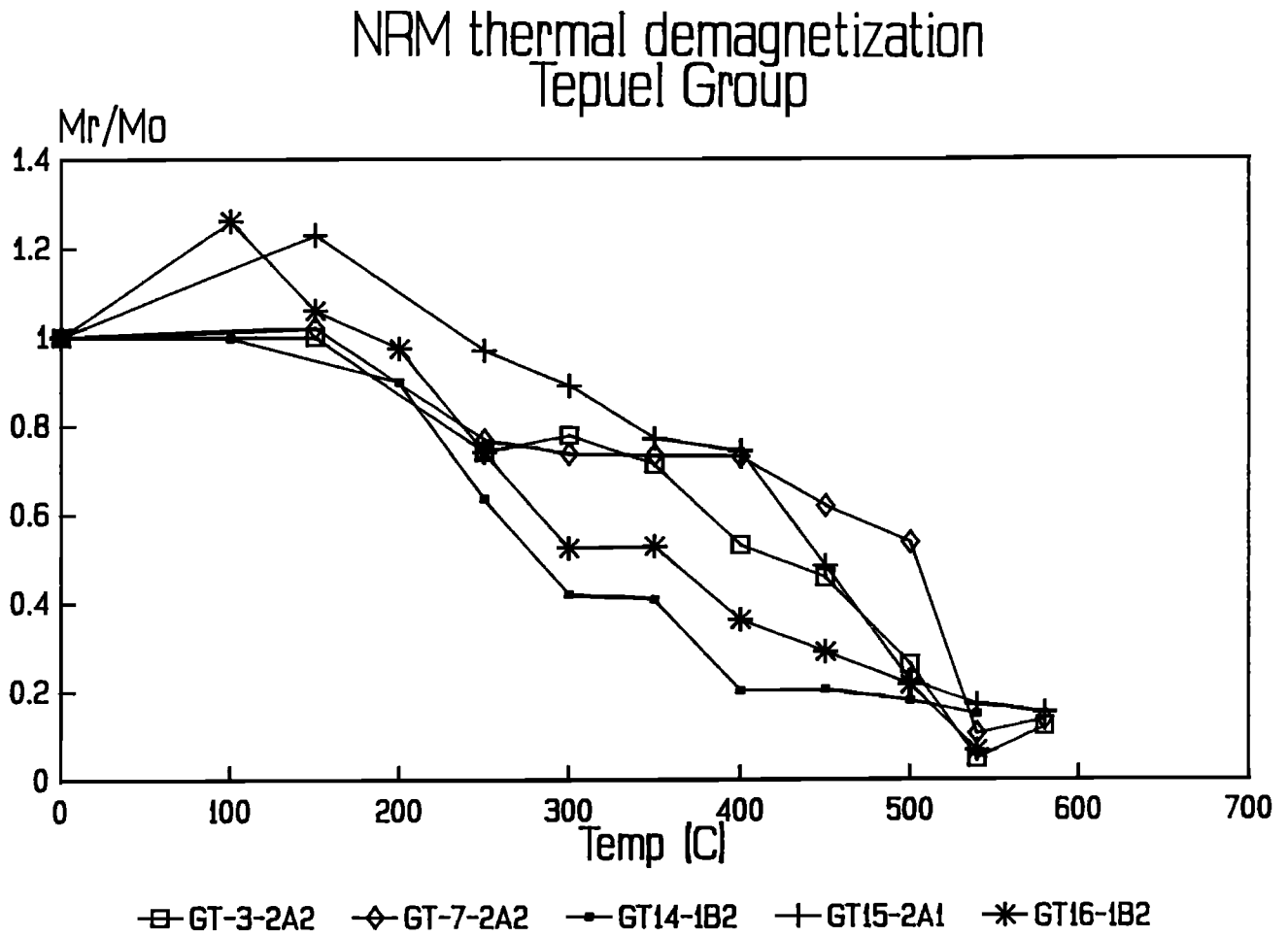


Figure 5. Normalized demagnetization curves for samples (see symbol legend) of the Tepuel Group. Note the wide spectra of blocking temperatures. Mo, intensity of initial remanence; Mr, intensity of remanence after each demagnetization step.

Table 1. Site Mean Direction for Component A from Sierra de Tepuel

Site	n	Remanence Direction In situ				Bedding Correction		Remanence Direction Corrected			
		Dec	Inc	k	$\alpha 95$	Strike	Dip	Dec.	Inc.	k	$\alpha 95$
GT-1	4	26.8	-45.8	24.5	18.9	69	29	13.0	-23.9	24.5	18.9
GT-2	1	343.3	-58.8	0.0	0.0	49	25	334.1	-35.2	0.0	0.0
GT-3	5	0.1	-48.0	17.4	18.8	49	25	348.7	-27.4	17.4	18.8
GT-4	10	5.7	-68.9	21.9	10.5	49	25	342.0	-47.8	21.9	10.5
GT-5	3	39.3	-54.4	431.9	5.9	49	25	12.0	-44.0	431.9	5.9
GT-6	6	17.3	-63.1	15.4	17.6	68	34	358.0	-32.9	15.4	17.6
GT-7	3	10.0	-74.7	33.6	21.6	68	34	348.9	-42.4	33.6	21.6
GT-8	4	20.7	-45.2	16.4	23.4	43	40	357.6	-21.9	16.4	23.4
GT-10	3	5.0	-40.4	15.3	32.7	43	40	350.8	-11.2	15.3	32.7
GT-11	2	18.8	-46.2	240.5	16.2	43	40	355.9	-21.7	240.5	16.2
GT-12	1	22.9	-49.3	0.0	0.0	43	40	355.9	-25.9	0.0	0.0
GT-13	3	1.1	-49.9	23.2	26.2	43	40	343.3	-18.0	23.2	26.2
Total											
	N=12	13.3	-54.5	35.7	7.4			353.4	-29.8	30.1	8.0

All remanence directions and bedding correction values are in degrees. N is number of sites considered in the mean, n is number of specimens per site; Dec, declination; Inc, inclination; and k and $\alpha 95$ are Fisherian statistical parameters. Direction of dip is 90° clockwise from strike.

Table 2. Sample Directions for Component B From Sierra de Tepuel and Characteristic Remanence From Las Salinas

Sample	Remanence Direction In situ				Bedding Correction		Remanence Direction Corrected			
	Dec	Inc	k	$\alpha 95$	Strike	Dip	Dec	Inc	k	$\alpha 95$
					Locality A, Sierra de Tepuel					
GT3-2A2	328.0	58.2			49	25	355.1	82.0		
GT3-2B1	358.6	29.9			49	25	13.5	47.2		
GT4-1A	70.3	69.3			49	25	106.2	52.5		
GT4-2A1	329.4	46.2			49	25	340.9	70.4		
GT4-3A2	357.5	50.3			49	25	31.1	65.3		
GT4-3B1	332.8	58.1			49	25	8.8	80.5		
GT4-3B2	0.5	47.3			49	25	30.6	61.7		
GT6-1C	0.5	52.6			68	34	62.8	76.5		
GT6-2B	222.2	72.9			68	34	180.5	46.1		
GT6-3B1	343.9	72.9			68	34	152.1	72.9		
GT7-2B	346.8	49.8			68	34	22.5	81.9		
GT8-4B	212.0	83.8			43	40	142.2	48.4		
GT9-2	2.5	53.0			43	40	68.1	59.7		
GT9-4A	6.5	33.0			43	40	40.3	47.6		
GT10-4B	332.0	45.3			43	40	30.1	76.4		
GT10-5A	321.4	79.3			43	40	129.8	60.6		
GT11-2	317.8	55.9			43	40	108.5	83.5		
GT11-3A	157.9	80.4			43	40	138.3	41.2		
GT11-4B	323.0	41.2			43	40	355.2	78.8		
GT12-1B	353.3	59.3			43	40	80.5	65.4		
GT12-2	15.2	63.5			43	40	89.6	55.0		
GT12-3	4.7	55.1			43	40	72.6	58.9		
GT12-4B	333.7	35.2			43	40	6.4	68.9		
Total (23)	346.6	59.6	12.8	8.8			75.4	74.5	10.9	9.6
					*Locality B, Las Salinas**					
GT14-1A1	197.1	54.0			254	28	244.2	71.0		
GT14-1A2	196.0	24.3			254	28	208.6	46.5		
GT14-2A1	211.3	32.1			254	28	231.6	47.6		
GT14-2A2	146.8	37.1			254	28	132.8	62.9		
GT14-2B1	174.5	37.7			254	28	183.9	64.8		
GT15-1A	124.1	44.6			254	28	92.4	61.2		
GT15-1B	62.0	59.6			254	28	28.9	45.4		
GT15-2A2	120.1	43.6			254	28	89.2	58.7		
GT15-2B	160.6	54.8			254	28	148.5	82.6		
GT15-2C	21.0	83.8			254	28	350.9	56.8		
GT15-3A1	161.1	70.0			254	28	351.1	81.9		
GT16-1A1	164.9	77.8			254	28	343.3	74.2		
GT16-1A2	232.2	49.3			254	28	267.2	51.5		
GT16-2B	223.4	51.9			254	28	263.9	57.4		
GT16-3A1	173.6	65.5			254	28	297.7	84.5		
GT16-3A2	96.9	60.7			254	28	45.8	59.3		
GT16-3B1	295.6	78.0			254	28	329.0	53.0		
GT17-2A2	117.8	54.6			254	28	70.0	65.2		
GT17-3A	145.5	64.3			254	28	50.2	81.3		
GT18-1A1	191.2	45.5			254	28	220.0	67.2		
GT18-1A2	162.6	19.4			254	28	162.0	47.4		
GT18-1B	157.8	61.5			254	28	80.1	87.0		
GT18-3A	28.6	-52.5			254	28	72.6	-64.7		
GT18-3B1	125.4	33.9			254	28	105.0	52.8		
GT18-3B2	181.2	52.5			254	28	215.9	76.7		
GT19-2A2	82.1	75.2			254	28	15.0	60.5		
GT19-2B	96.8	73.9			254	28	19.6	64.0		
GT19-3A1	116.2	67.8			254	28	37.1	69.5		
GT19-3A2	124.3	54.0			254	28	77.0	67.9		
GT19-3B1	101.3	31.9			254	28	81.2	40.5		
GT19-3B2	73.4	72.3			254	28	18.0	57.1		
GT20-2A1	103.1	58.4			284	21	73.1	52.4		
GT20-2A2	353.8	81.7			284	21	8.1	61.0		
GT21-1A1	176.9	69.4			284	21	92.7	83.9		

Table 2. (continued)

Sample	Remanence Direction In situ				Bedding Correction		Remanence Direction Corrected			
	Dec	Inc	k	$\alpha 95$	Strike	Dip	Dec	Inc	k	$\alpha 95$
GT21-1A2	151.0	60.3			284	21	108.5	70.2		
GT21-1B1	135.8	55.8			284	21	101.6	61.4		
GT21-1B2	162.7	68.3			284	21	92.6	78.7		
GT23-1A2	356.3	84.1			284	21	10.0	63.4		
GT23-2B1	125.2	58.1			284	21	89.7	59.4		
Total (39)	152.5	65.1	8.7	8.2			62.9	82.8	8.5	8.3
Total of A and B (62)	120.1	84.4	5.1	8.9			70.0	79.7	9.2	6.3

Abbreviations and units are same as in Table 1.

* Bedding correction for samples from locality B were done in the following two-step process: correction of the remanence directions and bedding attitudes for 30°fold plunging to N25°E (see text) and standard bedding correction by "unplunged" attitude.

remanence (IRM) studies of one specimen per site were carried out. The dc field was applied with a Molspin pulse magnetizer using incremented fields up to a peak value of 0.8 T. The IRM acquired in this peak field was then progressively destroyed by applying increasingly stronger fields in the opposite direction. Typical responses to these procedures (Figure 6) showed that saturation was largely achieved in low fields (0.1 to 0.4 T), other than the sample from site GT-20. The backfield coercivity (Hcr, the value of the reversed applied field at which the saturation magnetization in the opposite direction falls to zero) ranges from 36 to 50 mT. The direct field required to saturate (Hsat) and the backfield coercivity are thus consistent with the thermal demagnetization evidence that the dominant or sole ferromagnetic component is magnetite. In one site only (GT-20) is another ferromagnetic mineral significant, which was determined as goethite from both the thermal demagnetization curves and IRM behavior. Comparison of

Table 3. Site Mean Directions for Component B From Sierra de Tepuel and Characteristic Remanence From Las Salinas.

Site	n	Remanence Direction In Situ				Bedding Correction		Remanence Direction Corrected			
		Dec	Inc	k	$\alpha 95$	Strike	Dip	Dec	Inc	k	$\alpha 95$
Locality A, Sierra de Tepuel											
GT-3	2	347.1	45.0	10.7	86.2	49	25	10.4	64.8	10.7	86.2
GT-4	5	353.6	57.9	15.1	20.3	49	25	41.7	72.3	15.1	20.3
GT-6	3	337.8	75.0	10.1	41.1	68	34	158.2	71.0	10.1	41.1
GT-9	2	4.8	43.0	32.2	45.5	43	40	52.1	54.4	32.2	45.5
GT-10	2	329.8	62.4	11.3	82.9	43	40	102.7	74.6	11.3	82.9
GT-11	3	318.3	65.0	7.4	49.1	43	40	124.4	74.7	7.4	49.1
GT-12	4	353.4	54.4	24.1	19.1	43	40	68.9	65.2	24.1	19.1
Total N=	7	346.5	58.4	33.7	10.5			69.5	74.1	20.7	13.6
Locality B, Las Salinas											
GT-14	5	185.6	39.2	12.9	22.1	254	28	208.7	63.7	12.9	22.1
GT-15	6	122.7	64.5	11.0	21.2	254	28	52.8	71.6	11.0	21.2
GT-16	6	200.6	73.8	10.0	22.2	254	28	315.6	72.4	10.0	22.2
GT-17	2	129.6	60.2	46.2	37.6	254	28	69.6	73.4	46.2	37.6
GT-18	6	169.2	47.6	10.3	22.0	254	28	182.5	75.3	10.3	22.0
GT-19	6	103.1	63.7	19.1	15.7	254	28	48.1	63.2	19.1	15.7
GT-20	2	87.1	75.0	10.7	85.9	284	21	45.8	61.0	10.7	85.9
GT-21	4	153.7	64.3	65.1	11.5	284	21	102.2	73.7	65.1	11.5
GT-23	2	115.5	75.6	10.3	88.3	284	21	53.9	67.2	10.3	88.3
Total N=	9	147.4	66.8	16.7	13.0			63.6	79.8	16.3	13.2
All sites N=	16	56.0	85.1	6.5	15.7			69.6	77.5	19.6	8.5

Abbreviations and units are same as Table 1. Bedding correction for Locality B is same as Table 2.

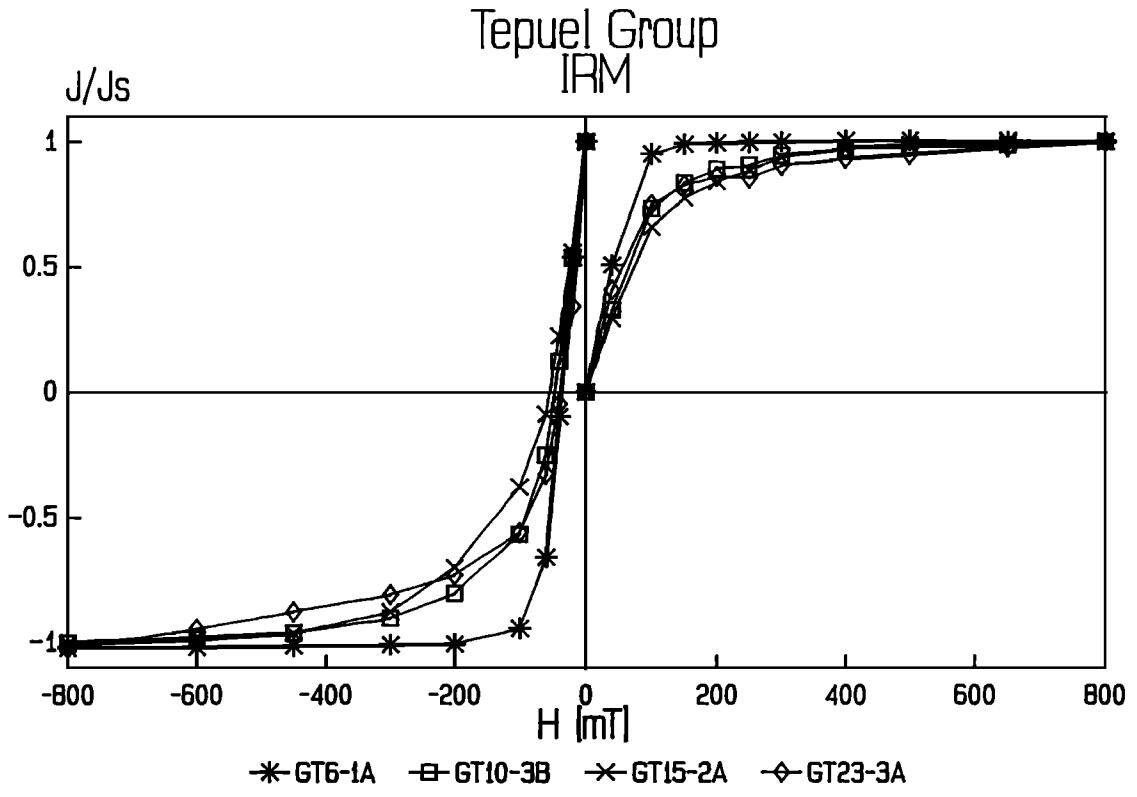


Figure 6. Normalized isothermal remanent magnetization (IRM) forward and backward acquisition curves for samples (see symbol legend) of the Tepuel Group. J, IRM intensity after each magnetization step; Js, saturation IRM intensity.

the observed Hcr values with those for magnetite grains of known size [*Dankers*, 1981] suggests that the dominant grain sizes in these sediments are smaller than 20 μm, implying pseudo-single-domain (PSD) behavior [*Levi and Merrill*, 1978]. The fact that MD grains were suspected in many sites during natural remanent magnetization (NRM) demagnetization suggests that a broad range of grain sizes is probably present in many sites.

Thermal demagnetization of the peak IRM (0.8 T) largely erased it at 580°C and erased it entirely by 610°C (Figure 7) - with only a small tail of remanences remaining between both steps. This strongly suggests that the IRM is carried by magnetite, with the tail probably being of hematite produced during the heating process, as indicated by important chemical changes when heating over 500°C. The disturbance of the NRM by heating was evidenced by comparison of the IRM curves for the same specimen before and after heating to 610°C (Figure 8). Apparently, during heating both low coercivity (magnetite?) and high coercivity (hematite?) minerals are produced, as suggested by the fivefold increase in the remanence intensity and the lack of saturation at 0.8 T.

Petrographic Properties

Fifteen polished thin sections were prepared from representative lithologies of the Tepuel Group and examined using transmitted and reflected light microscopy. All of the sections show small amounts of secondary iron hydroxides, mainly after sulphides (Figure 9a). Most of these secondary minerals were extremely fine grained (< 1 μm). Their paragenesis is difficult to determine because of their fine-grained nature, but they are almost certainly postdiagenesis and most likely to be associated with recent weathering. They are unlikely to contribute significantly to the observed remanence because of their grain sizes but may give rise to small, random magnetic moments.

In only a few samples could magnetite grains be identified in reflected light. These were typically pale brown in color and had an alteration selvage which could not be resolved optically (Figure 9b). Their grain shape and textural association clearly indicate that this magnetite is of detrital origin.

Interpretation and Discussion

The rock magnetic and petrographic evidence suggests that one or more of the observed components of remanence may be associated with detrital magnetite and hence record the direction of the geomagnetic field while the sediments were forming. There is also petrographic evidence for goethite that is likely to be associated with a secondary weathering component, but its presence seemed not to be relevant in the remanence of most sites.

In situ site mean directions of component A from Sierra de Tepuel (Table 1 and Figure 10) show consistency with the

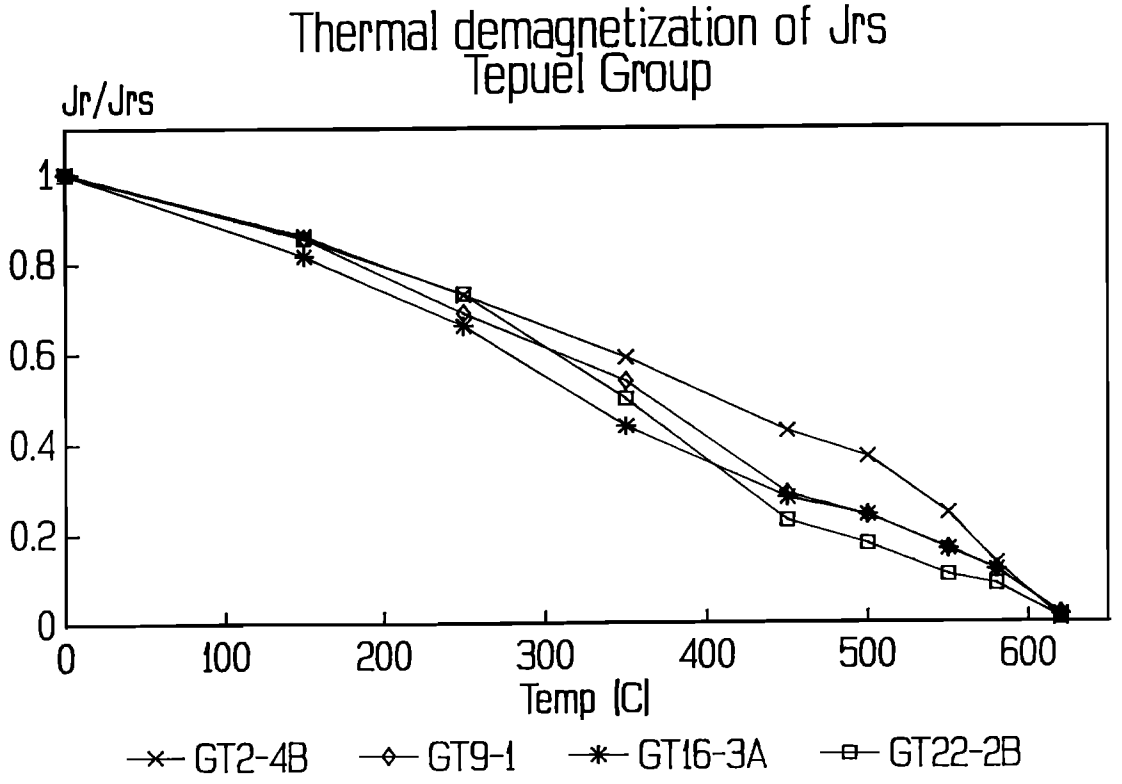


Figure 7. Thermal demagnetization of saturation IRM (0.8 T) for samples (see symbol legend) of the Tepuel Group. Jr, IRM intensity after each thermal demagnetization step; Jrs, saturation IRM intensity.

already established post Early Permian directions for South America [*Irving and Irving*, 1982], while correction for tectonic tilt not only moves them away from any expected late Paleozoic to Mesozoic magnetization direction but also increases their scatter. This demonstrates that component A has a posttectonic origin of post-Carboniferous age. However, a Permian age does not seem likely because of the exclusive normal polarity, nor does a late Tertiary or Quaternary age because the present dipole direction is outside the 95% confidence circle about the component A mean direction. Thus the age of the secondary component A can be confined to the Triassic-early Tertiary. The paleomagnetic pole computed from the in situ mean direction (163.7°E, 76.6°S, minor and major semi-axes of confidence $\delta p = 7.3°$, $\delta m = 10.4°$) is close to Early and Middle Jurassic poles from Central Patagonia [*Mena*, 1990; *Vizán et al.*, 1990; *Rapalini and Vilas*, 1991], suggesting that component A could be a thermoviscous magnetization, possibly acquired during the Early-Middle Jurassic intrusion of the gabbroic dykes of the "Cresta de los Bosques Complex" [*Page*, 1984]

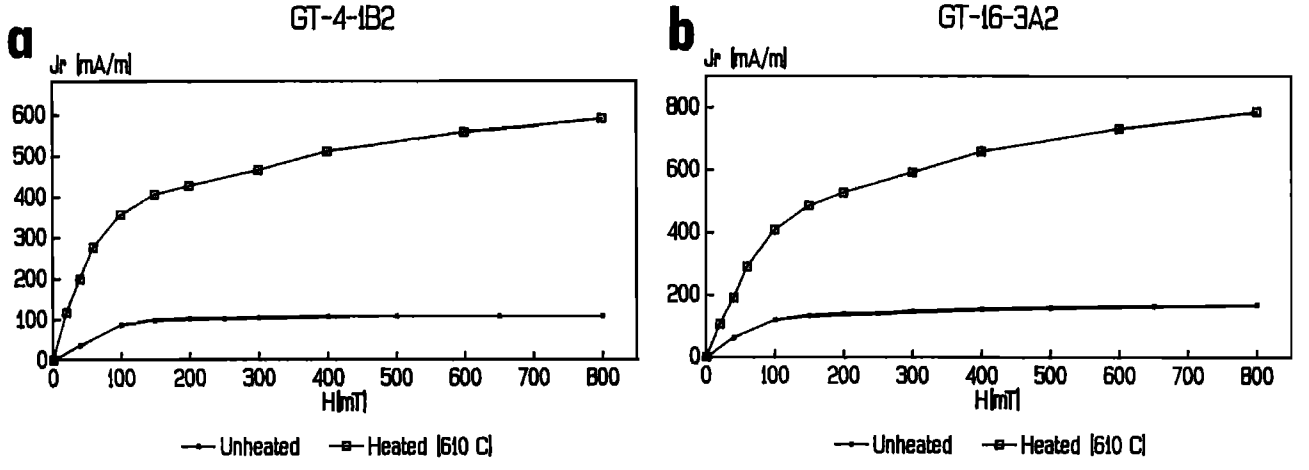


Figure 8. IRM acquisition curves for specimens of the Tepuel Group before and after being heated at 610°C.

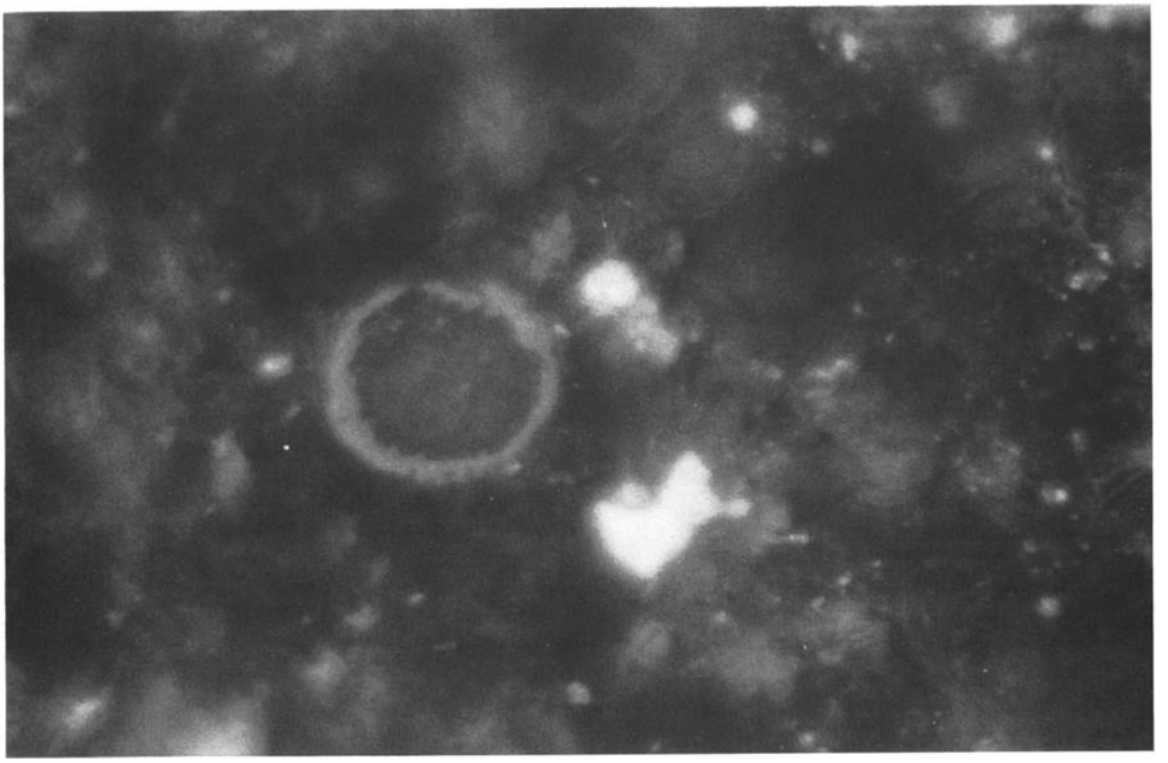


Figure 9a. Polished thin section of a specimen of the Tepuel Group showing secondary iron hydroxides after sulphides (pyrite framboid?).

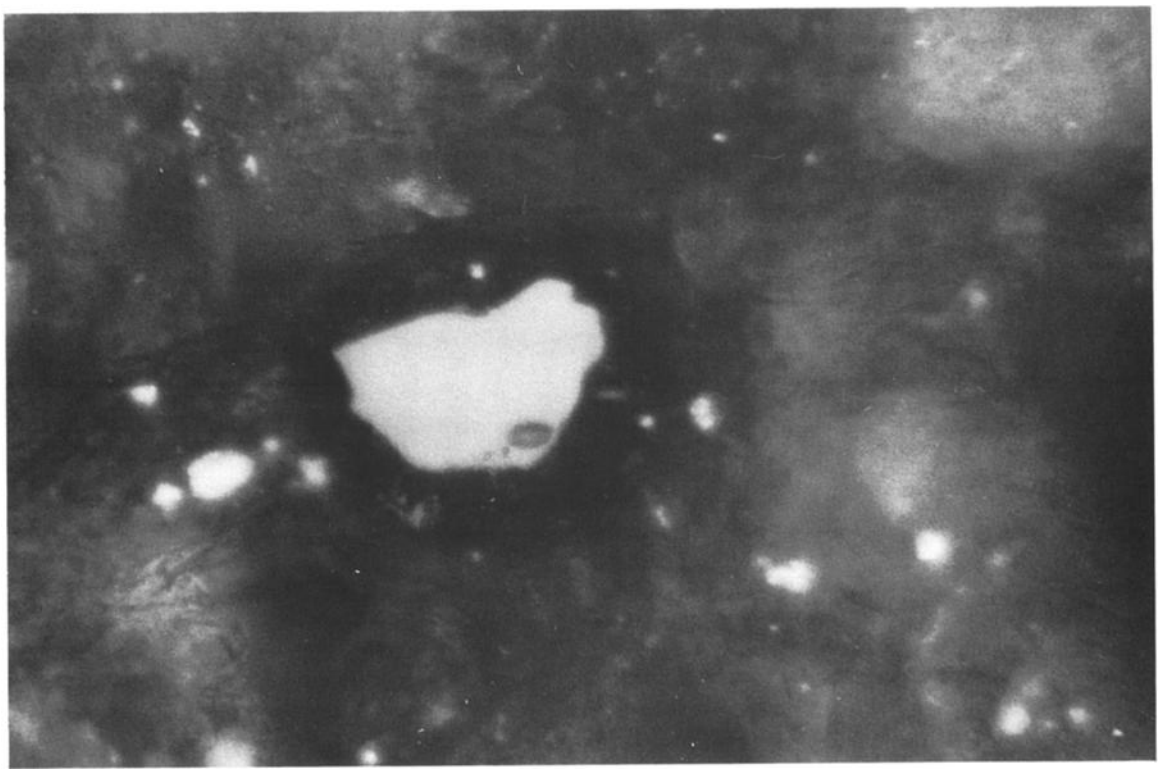


Figure 9b. Another specimen of the Tepuel Group showing a large detrital grain of magnetite, through reflected light.

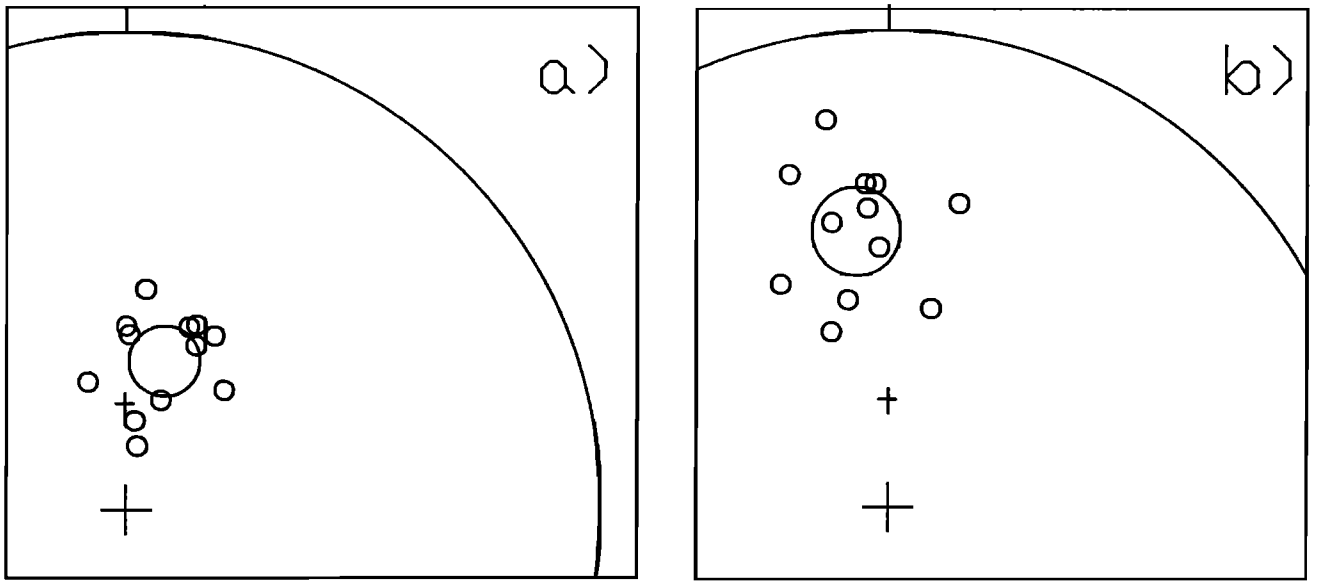


Figure 10. Site mean directions of component A from the Tepuel Group at Sierra de Tepuel (a) in situ, and (b) after bedding correction.

in Sierra de Tepuel. Component A was not found in Las Salinas (locality B) where these gabbroic intrusives are also absent. However, owing to the very similar South American pole positions throughout the Mesozoic and Cenozoic, a more recent origin related to either Cretaceous magmatic activity [*Franchi and Page*, 1980] or to the uplift of the Andean Cordillera in the Tertiary cannot be ruled out.

Component B of Sierra de Tepuel and the characteristic remanence of Las Salinas both belong to rocks stratigraphically similar and have comparable magnetic properties (Curie points, coercivities, and polarity of the magnetizations). In situ sample directions from both localities form two separate groups but are mutually consistent after correction for tectonic tilt (Table 2 and Figures 11a and 11b). The same result is obtained if the site mean directions are used (Table 3 and Figure 11c). A composite bedding correction was performed in the case of samples (sites) from locality B, accounting for the 30° plunge of the fold axis toward N25°E in this locality. A first step consisted of the "unplunging" of remanence directions and bedding attitudes, followed by a normal bedding correction of the remanences according to the unplunged bedding attitudes. However, owing to the low angle of tilting at this locality, this composite structural correction does not yield significantly different corrected remanence directions than the standard bedding correction. The tilt test [*Graham*, 1949] is positive, indicating a pretectonic origin for the remanence. The application of the traditional McElhinny's [1964] statistical test yields a $k2/k1$ (Fisher's statistical parameter *kappa* after and before tectonic correction, respectively) value of 3.02 (number of sites (N)=16) which indicates a positive test at 99% confidence level. The new *McFadden* [1990] fold test was also applied, yielding the following results: 1) on a site level the SCOS (*McFadden*'s parameter for evaluation of the significance of the test) statistic is significant at 99% (12.525 versus 6.516) before untilting but is not significant at 95% (3.499 versus 4.658) after correction; stepwise untilting gives its minimum SCOS value at 83% of correction; and 2) on a sample level the SCOS statistic is significant at 99% (33.48 versus 12.951) in situ, but not significant at 95% (1.506 versus 9.159) after bedding correction; partial untilting gives its minimum SCOS value at 96% of bedding correction. These results strongly argue for a pretectonic magnetization, confirming the visual grouping of directions after bedding correction and McElhinny's test. The minimum SCOS values at partial untilting are not significantly different than those at 100%. A suspicion of syntectonic magnetization due to the marginally better statistical values at bedding corrections a few percentages less than 100% is also unlikely, considering that localities A and B are situated in different tilted blocks. If *Gonzalez*'s [1972] suggestion of an Early Permian age for the tectonic event that folded the rocks at Las Salinas is accepted, then the age of the magnetization in both localities should be pre-Early Permian. However, a more conservative interpretation of the tilt test would only constrain the age of the magnetization as pre-Cretaceous, the youngest possible age of tectonic tilting at Sierra de Tepuel [*Suero*, 1948; *Spikermann*, 1978). All samples, except one, presented reversed polarity, consistent with a Carboniferous age [*Palmer et al.*, 1985]. The single sample with normal polarity at site 18 (Table 2) could represent magnetization during an excursion or short reversal of the late Paleozoic magnetic field.

The paleomagnetic, rock magnetic, and petrographic studies suggest that the magnetic carrier in these rocks is detrital magnetite; the magnetization is thus most likely to be depositional and/or postdepositional (detrital magnetization [*Tarling*, 1983]). Therefore a primary origin of Namurian to Westphalian age is assumed. A paleomagnetic pole (TP),

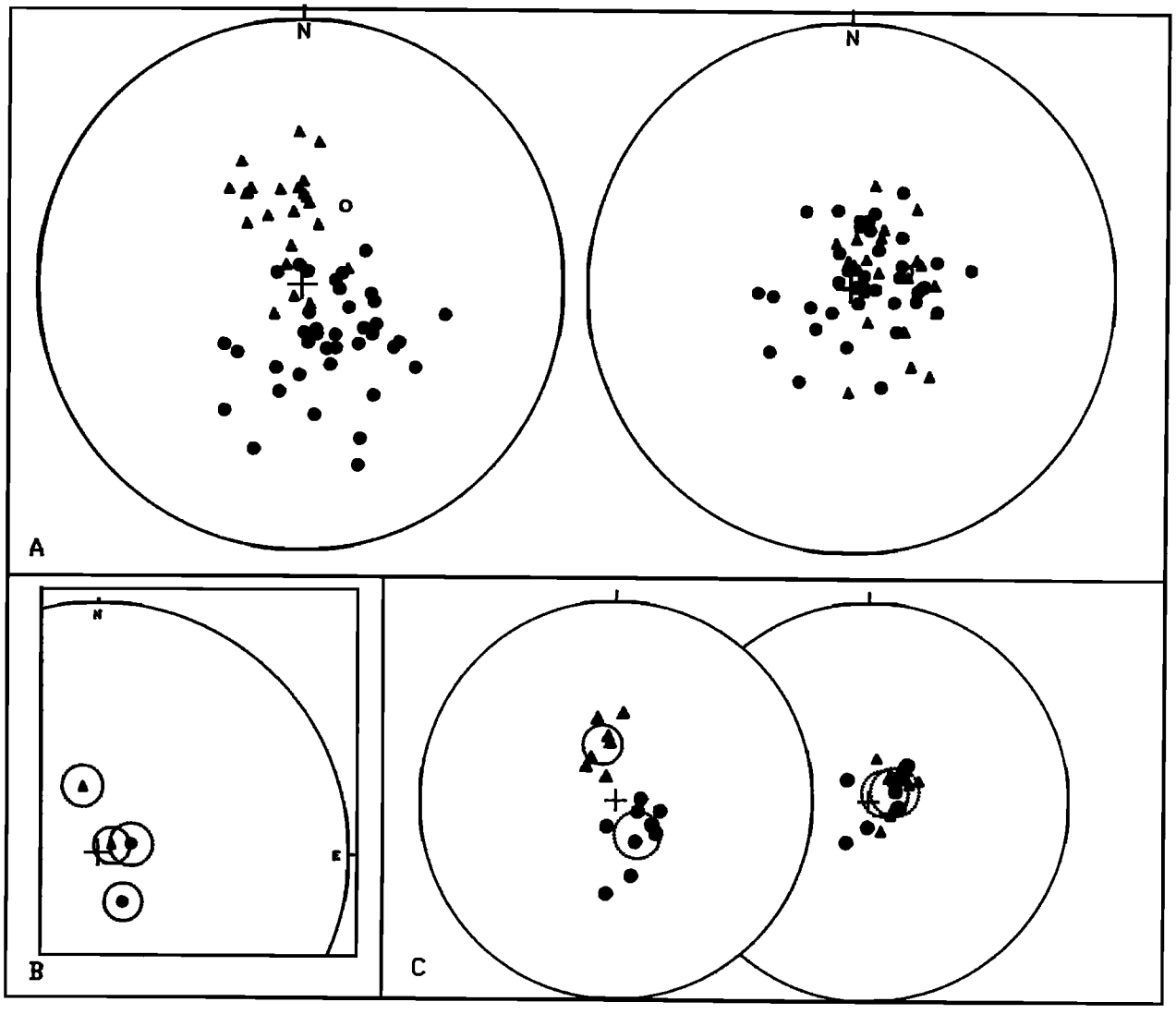


Figure 11(a). Sample directions of component B from Sierra de Tepuel (triangles) and the characteristic remanence from Las Salinas (circles) for (left) and (right) after bedding corrections. (b) Averaged sample directions and respective 95% confidence circles ($\alpha 95$) for localities A (triangles) and B (circles) before and after applying the tectonic corrections. Note the overlap of the confidence circles after tilt correction. (c) Site mean directions of component B (triangles) and the Chrm from locality B (circles) for (left) in situ and (right) tilt corrected. Large dotted circles in Figures 11b and 11c indicate the $\alpha 95$ of the averaged direction for each locality.

based on the mean directions for all sites (Table 2), is at 316.1°E, 31.7°S (N=16, δp=15.0°, δm=16.0°). Owing to the low within-site consistency of directions, an alternative paleomagnetic pole was computed on a sample basis, 312.3°E, 34.2°S (number of samples (n)=62, δp=11.5°, δm=12.0°). In whichever manner one computes the Tepuel Group pole (TP) meets most modern reliability criteria. This means that the mean pole position can be used for tectonic and paleogeographic interpretations.

The new pole, in present geography (Figure 12), is east of the Brazilian coast and well displaced from the established, reliable latest Carboniferous-Permian poles from the Paganzo and Paraná basins in South America [*Creer*, 1965; *Embleton*, 1970; *Sinito et al.*, 1979; *Valencio et al.*, 1975; *Pascholati and Pacca*, 1976], as well as from the mean South American Permian and Triassic paleomagnetic poles [*Rapalini et al.*, 1993]. TP does, however, coincide almost exactly with the only other published mid-Carboniferous pole for South America, that of the Taiguati Formation (TG) [*Creer*, 1965]. Unfortunately, the original treatment and analysis of the Taiguati data no longer fulfill modern criteria, but neither is there evidence for specific unreliability. The Tepuel Group and the Taiguati Formation include the *Levipustula Levis* zone [*Lopez Gamundi*, 1989], which indicates a Namurian-early Westphalian age for both sequences. TP confirms the mid-Carboniferous pole position for South America previously determined by *Creer* [1965], suggesting that this continent was then situated in polar

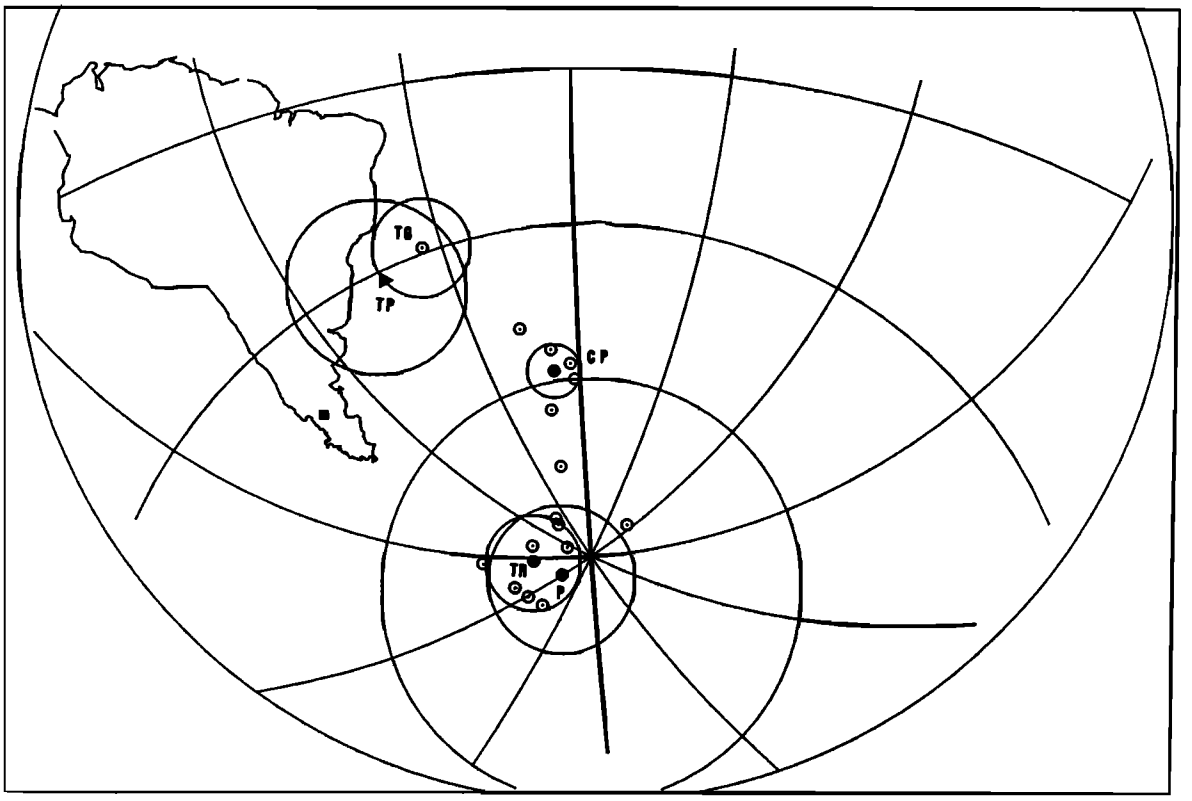


Figure 12. Location of the Namurian-Westphalian paleomagnetic pole for the Tepuel Group (TP) and its oval of 95% confidence. TG is the paleomagnetic pole of the Taiguati Formation [*Creer*, 1965] and its 95% confidence circle (A95) on a sample basis. Solid circles represent South American mean paleomagnetic poles for the Late Carboniferous-Early Permian (CP), Late Permian (P), and Triassic (TR) computed from reliable paleomagnetic poles (open circles) according to a recent selection by *Rapalini et al.* [1993].

Table 4. Selected Paleomagnetic Poles for the Definition of the Gondwana Apparent Polar Wander Path Shown in Figure 13

Geologic Unit	Age	Paleomagnetic Pole			Q	Reference
		Latitude, deg	Longitude, deg	A_{95},$\delta p/\delta m$, deg		
AIR, Nigerian Ring Complex, Africa	S l	-43.4	8.6	6.2	6	1
SR, Snowy River Volcanics, Australia	D l	-74.3	222.7	14.5/10.9	7	2
CV, Comerong Volcanics, Australia	D m-u	-76.9	330.7	7.2	6	3
GH, Gilif Hills Volcanics, NE Africa*	D m	26.0	12.0	11.0	6	4
WP, Worange Point Formation, Australia	D u	-70.8	19.7	7.1	6	5
HG, Hervey Group, Australia	D u-C l	-54.4	24.1	16.2/8.4	5	6
CB, Canning Basin Limestone, Australia	D u-C l	-49.1	38.0	7.8	6	7
TP, Tepuel Group, Patagonia	C m	-31.7	316.1	16.0/15.0	6	8
DV, Dwyka Varves, South Africa	C m	-26.5	26.5	10.5	5	9
C-P, mean PP, South America and Africa	C u-P l	-24.5	59.0	3.8	-	10
P-T, mean PP, South America and Africa	P u-Tr l	-57.7	72.5	8.3	-	11

Abbreviations are A_{95}, radius of 95% confidence circle, $\delta p/\delta m$, major and minor axes of 95% confidence oval; Q, assigned reliability index according to *Van der Voo*'s [1990] criteria (maximum 7); S, Silurian; D, Devonian; C, Carboniferous; P, Permian; Tr, Triassic; l, lower; m, middle; and u, upper. References are 1, *Hargraves et al.* [1987]; 2, *Schmidt et al.* [1987]; 3, *Schmidt et al.* [1986]; 4, *Bachtadse and Briden* [1991]; 5, *Thrupp et al.* [1991]; 6, *Li et al.* [1988]; 7, *Hurley and Van der Voo* [1987]; 8, this paper; 9, *McElhinny and Opdyke* [1968]; 10, Selected western Gondwana Late Carboniferous - Early Permian poles [*Rapalini et al.*, 1993] and paleomagnetic pole from ElAdeb Larache Formation [*Henry et al.*, 1992]; and 11, Selected western Gondwana Late Permian - Early Triassic poles [*Rapalini et al.*, 1993].

* Paleomagnetic pole not used in the definition of the apparent polar wander path.

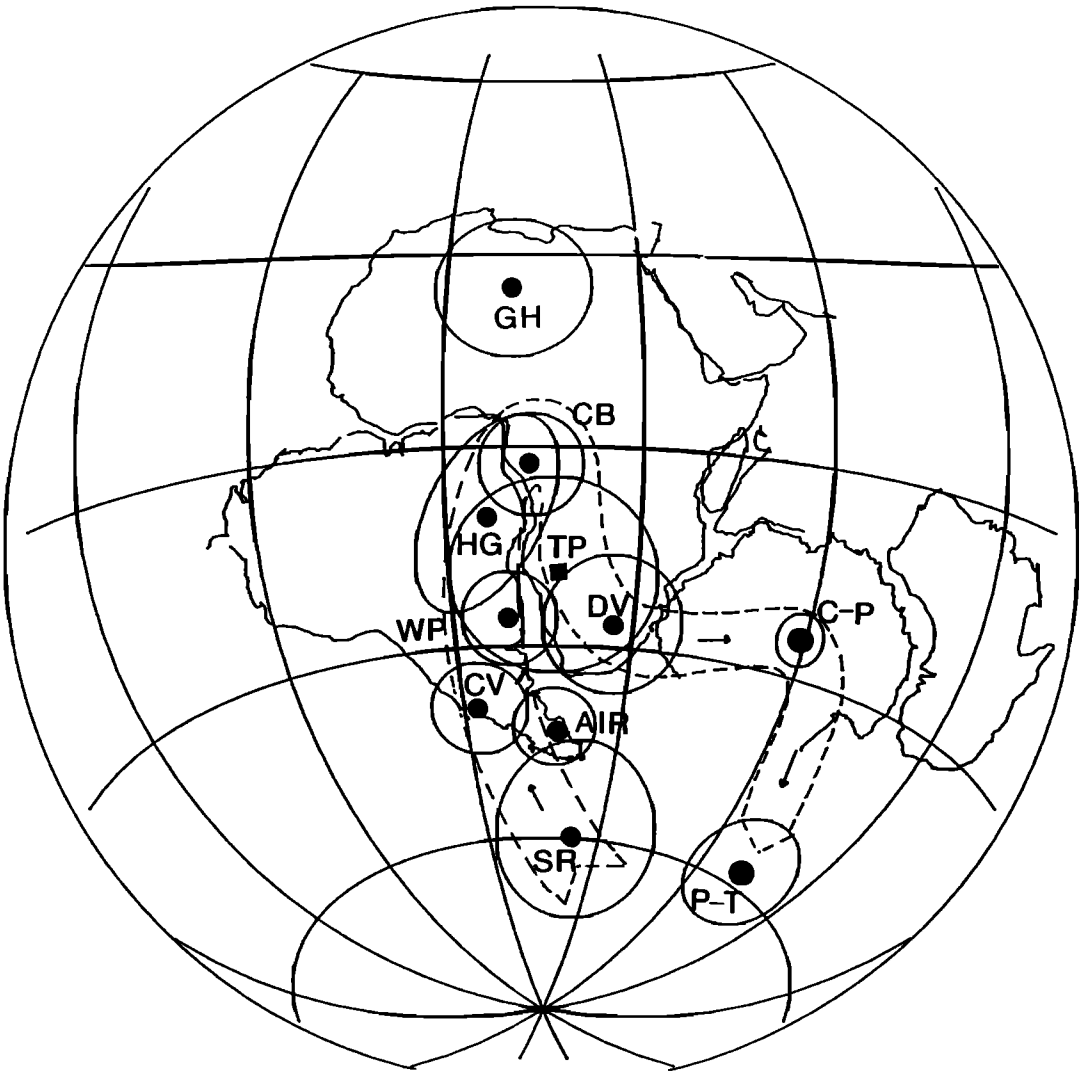


Figure 13. Silurian to Permian reliable paleomagnetic poles for Gondwana and its schematic apparent polar wander path (dashed-line path). Poles of Silurian to Middle Carboniferous age are represented as solid circles and with their oval or circle of 95% confidence. Larger circles correspond to Late Carboniferous - Early Permian (C-P) and Late Permian - Early Triassic (P-T) mean poles for western Gondwana. Square (TP) shows position of the paleomagnetic pole for the Tepuel Group. Gondwana reconstruction is based on that by *Lottes and Rowley* [1990]. Other poles are AIR, Nigerian AIR Ring complex [*Hargraves et al.*, 1987]; SR, Snowy River volcanics [*Schmidt et al.*, 1987]; CV, Comerong volcanics [*Schmidt et al.*, 1986]; GH, Gilif Hills volcanics [*Bachtadse and Briden*, 1991]; WP, Worange Point Formation [*Thrupp et al.*, 1991]; HG, Hervey Group [*Li et al.*, 1988]; CB, Canning Basin [*Hurley and Van der Voo*, 1987]; DV, Dwyka varves [*McElhinny and Opdyke*, 1968].

latitudes, so that the vast glacial deposits in Brazil and Argentina [*Crowell and Frakes*, 1975; *Gonzalez*, 1990; *Lopez Gamundi et al.*, 1992], of which the glacial horizons of the Tepuel Group are examples, were probably the product of a polar ice sheet [*Gonzalez Bonorino*, 1992]. At some time during the Westphalian, South America moved rapidly into much lower latitudes, as evidenced by the important amelioration of the climatic conditions [*Gonzalez*, 1990; *Lopez Gamundi et al.*, 1992].

The position of TP contributes to define the middle to late Paleozoic section of the Gondwanan APWP. This is still a matter of controversy, as two completely different paths seem possible for the Silurian-Devonian. One of them [*Li et al.*, 1988; *Thrupp et al.*, 1991] implies a long Devonian APWP, with a path going from southern South America in the Early Devonian to central western Africa in the Early Carboniferous. This is based on several high-quality paleomagnetic poles (Table 4), although most of them come from the Tasman Fold Belt (Figure 13). An alternative path [*Bachtadse and Briden*, 1991] resembles the *X* path of *Morel and Irving* [1978] and involves a slow movement of the pole from North Africa in the early Paleozoic to central Africa by the end of the Devonian. Despite the fact that most reliable data tend to favor the previous path, a recent high-quality pole from volcanic rocks and tuffs of Middle Devonian age from the Saharan Craton (GH, Table 4) lies in northern

Africa and is more consistent with the second alternative. However, both alternatives coincide in an Early Carboniferous pole position on central western Africa. When plotted in a Gondwana reconstruction [*Lottes and Rowley*, 1990] the new TP pole lies on south-western Africa close to the Dwyka Varves pole (DV) [*McElhinny and Opdyke*, 1968], which is generally considered as being of Middle Carboniferous age, although its age is not precisely determined. The CB, TP, and DV poles (Figure 13) now define a reliable Early to Middle Carboniferous APWP for Gondwana that implies a displacement of the pole from central western Africa to southern Africa between the Tournaisian (350 Ma) and the Namurian (325 Ma) at moderate drift rates. Apparently, the pole remained in southern Africa only until the Westphalian, when Gondwana experienced a fast movement in a different direction, as evidenced by the position of the Late Carboniferous poles over Antarctica (C-P, Figure 13). A recent Moscovian (late Westphalian) paleomagnetic pole from the Sahara Craton [*Henry et al.*, 1992], consistent with other Late Carboniferous poles from Gondwana, indicates that this fast displacement had already ended by that time (310-300 Ma). A further change in drift direction then occurred during the Early Permian, as indicated by a substantially different position of the Late Permian poles [P-T, Figure 13).

The position of TP also has important implications for the Paleozoic tectonic evolution of both southern South America and southern Africa. A recent paleomagnetic study on Silurian-Devonian rocks from NE Patagonia [*Rapalini and Vilas*, 1991] yielded an Early Devonian pole, in close agreement with the poles of the Early Devonian Snowy River volcanics (SR) and the Early Silurian Nigerian Ring complexes (AIR), suggesting no latitudinal displacements between Patagonia and Gondwana since the Early Devonian. However, this interpretation was hampered by the preliminary nature of the paleomagnetic study and the general uncertainties involving the mid-Paleozoic Gondwana path. The observed consistent position of TP on the Gondwana APWP confirms that by at least the Middle Carboniferous, Patagonia was already part of South America (within the paleomagnetic errors) and that little or no relative displacement between them has since occurred.

Conclusions

The new Patagonian pole is well established following paleomagnetic, rock magnetic and petrographic studies on mid-Carboniferous Tepuel Group samples from two localities (Sierra de Tepuel and Las Salinas) in western central Patagonia. This pole is based on a pretectonic (pre-Early Permian?) remanent magnetization that is considered to be primary, i.e., Namurian-early Westphalian, although it has been partially overprinted in Sierra de Tepuel by a secondary component apparently related to Mesozoic intrusions. The agreement of this mid-Carboniferous pole with other Gondwanan data confirms that the Patagonian block did not experience important relative displacements with respect to South America since this time. Thus this new pole also assists the definition of the Carboniferous section of the Gondwana APWP which moved from central Africa in the Early Carboniferous to southern Africa by the early Westphalian. It also suggests rapid relative polar motions in the late Westphalian.

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